

A Lightweight Small Object Detection Framework Based on Enhanced BiFPN and Attention Mechanisms

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Abstract

In this paper, we proposed an algorithm based on YOLO11n for a further improvement in the task of small object detection. First, we enhance the feature extraction capability by introducing a dynamic snake convolutional (DSConv), which significantly improves the ability to capture small object features of network. Then, we designed a new network structure named EBPB (Bi-directional Feature Pyramid Network with Efficient Multi-Scale Attention). This structure is aimed at extracting deeper and more informative features of small objects. Furthermore, we replace the original CIOU in YOLO with WIOU to address the limitations of the loss function and improving the model's feature learning capability. We name our model as ESW-YOLO (Enhanced YOLO by EBPB, Dynamic Snake Convolution, and WIOU), which achieves a 20.5% improvement in mAP50 over YOLO11n under the same parameter scale.

Keywords: Small Object Detection, YOLO11, Dynamic Snake Convolution, Small Object Loss Function

1. Introduction

Performance optimization for SOD (small object detection) has been a central challenge in object detection tasks for a long time. The performance of detection declines sharply when the pixel area occupied by objects in an image is 1024 or less, which is the main reason that mainstream algorithms struggle to achieve a further breakthrough. Contributing factors include the difficulty of feature extraction for small targets due to the influence of the loss function;

Since the success of AlexNet in 2012 [6], object detection algorithms with deep learning have become the mainstream method in this field. The current object detect methods can be broadly categorized into two main types: two-stage algorithms, such as the R-CNN family [4, 3, 15], and one-stage algorithms, that are represented by YOLO [12, 13, 14, 1, 7, 19, 20, 18] and SSD [8, 5]. To face the challenges in small object detection, researchers have proposed numerous improvement strategies from various perspectives, enriching and advancing the technical landscape of this subfield gradually. The current methods to improving SOD performance can be classified into six categories [2]: sample-oriented methods, scale-aware methods, attention-based methods, feature imitation methods, context modeling methods, and focus-and-detection methods.

To further enhance the small object detection performance of YOLO, we propose an architecture named ESW, incorporating Deformable convolutional kernels and a more advanced loss function specifically tailored for small objects. The main contributions of this work are as follows:

1. Incorporate Dynamic Snake Convolution [11] into the P3 layer of the backbone network to enhance deep feature extraction for small object images.

2. Propose the ESW architecture, inspired by the feature fusion strategy of BiFPN [16], which integrates iRMB [22] and EMA [10] attention mechanisms to construct a more efficient and robust feature extraction model.

3. Replace the CIOU loss with WIOU [17] in YOLOv11 to further improve detection performance, particularly for small objects.

2. Related works

2.1. YOLO11 Method

As a one-stage object detection framework, YOLO has been widely adopted in real-world industrial applications. Since its first appearance in 2016, YOLO has evolved significantly, with the latest iteration reaching 12 version.

Based on the YOLO methods, Gang Wang et al. [21] has proposed a model named UAV-YOLOv8 to enhance accuracy in remote sensing small object detection tasks. This model replaces CIOU loss with WIOU (Wise-IoU) v3, incorporates the BiFormer attention mechanism into the backbone, and introduces a feature processing module called the FFNB (Focal FasterNet Block). Yin Zhang et al. [23] has developed an efficient detector named FFCA-YOLO by integrating feature enhancement, fusion, and context-awareness strategies. FFCA-YOLO comprises three innovative, lightweight, and plug-and-play modules: Element feature enhancement module, FFM (Feature Fusion Module), and SCAM (Spatial Context-Aware Module). Haitong Lou et al. [9] has introduced DC-YOLOv8, which integrates the MDB module for down sampling and proposes the DC module (a network structure composed of stacked depth wise separable and standard convolutions) to improve the

C2f module in the YOLOv8 architecture. Additionally, the feature fusion strategy is optimized to enhance small object detection, achieving performance gains across tasks of all object sizes.

Among the various versions, YOLOv5, YOLOv8, and YOLO11 stand out for their representativeness and practical engineering value. The improvements proposed in this paper are based on YOLO11.

3. presented method

3.1. Dynamic snake convolution

Traditional convolutional kernels define sampling locations using a fixed pattern.

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n) \quad (1)$$

For a 3×3 kernel, let p_0 represent the center coordinate of the kernel, and let R denote the set of nine positions centered around the origin, and $P_n \in R$ corresponds to the coordinate of a sampling point relative to the center point. The associated weight at each location is denoted by w . In deformable convolution, an additional offset ΔP_n is added to each point, enabling the kernel to adapt its shape and better align with the spatial structure of the target.

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n + \Delta p_n) \quad (2)$$

This enables the convolutional kernel to deform, and allow it to adapt its shape dynamically during feature extraction.

Building upon traditional deformable convolution, Dynamic Snake Convolution introduces an iterative offset generation mechanism that imitate the movement of a snake's head guiding its body. In this approach, positions of convolution are determined based on the previous one, with the direction of head is selected randomly to simulate a swinging motion. For a standard 3×3 convolutional kernel, the offset Δp_n for each sampling points p_n are accumulated as follows:

$$\Delta p_n = \Delta p_{n-1} + \delta_n \quad (3)$$

Here, δ_n represents the incremental offset at the current step. Assuming the initial sampling point is p_0 , the position of the n th sampling point p_n is computed by the following equation:

$$p_n = p_0 + \sum_{k=1}^n \Delta p_k \quad (4)$$

The offset Δp_n is obtained by a convolution operation applied to the feature map.

$$\Delta p = \text{Conv}(X, W_{\text{offset}}) \quad (5)$$

In this equation, X represents the original feature map with dimensions $[B, C, H, W]$, where B is the batch size, C is the number of channels, H and W denote the spatial dimensions, and W_{offset} is the convolution parameters. The equation generate an offset tensor Δp_n with dimensions $[B, 2N, H, W]$, where $N = k \times k$ (e.g., $N = 9$ for a 3×3 kernel). The tensor Δp encodes the offsets both in the x and y directions for each sampling point.

Since the values of Δp_n are often fractional, the feature value at that point is approximated by using bilinear interpolation. The feature value is computed by performing bilinear interpolation with the four points that neighboring, then computed using the following formula.

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n) \quad (6)$$

Here, $w(p_n)$ represents the weights of $p_0 + p_n$, is calculated by bilinear interpolation, ensuring a smooth interpolation of the sampled value.

3.2. EBP network

3.2.1. BiFPN

BiFPN (Bidirectional Feature Pyramid Network), preserves both top-down (high-level semantics to low-level details) and bottom-up (low-level details to high-level semantics) pathways and forming a closed-loop information flow. It eliminates nodes with only a single input edge so that reduce redundant computation. A comparison between FPN, PAN, and BiFPN is illustrated in the Figure 1.

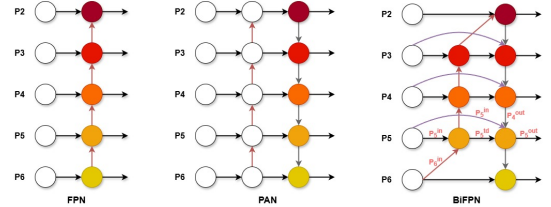


Figure 1: Comparison of FPN, PAN, and BiFPN.

Since the feature with different resolution contributions to the fused output are inherently unequal. Simply summing or stacking them is therefore suboptimal. BiFPN adopts a Fast Normalized Fusion approach to better handle this imbalance.

$$P_5^{td} = \text{Conv} \left(\frac{w_1 \cdot P_5^{in} + w_2 \cdot \text{Resize}(P_6^{in})}{w_1 + w_2 + \epsilon} \right) \quad (7)$$

$$P_5^{out} = \text{Conv} \left(\frac{w'_1 \cdot P_5^{in} + w'_2 \cdot P_5^{td} + w'_3 \cdot \text{Resize}(P_4^{out})}{w'_1 + w'_2 + w'_3 + \epsilon} \right) \quad (8)$$

The Resize operation includes down sampling to ensure all input feature maps share the same resolution, while w represents the weight assigned to each contributing feature map.

3.2.2. EMA attention mechanism

EMA (Efficient Multi-scale Attention) is an innovative and efficient multi-scale attention mechanism. To preserve information within each channel while reducing computational cost, EMA reshapes part of the channel dimension into the batch dimension and divides the channels into multiple sub-feature groups. Unlike traditional channel attention mechanisms that model inter-channel relationships through dimensionality reduction, EMA maintains the integrity of channel information without introducing additional parameters. It also segments the batch into several sub-feature groups, ensuring that spatial semantic features are distributed within each group evenly. In addition to generating channel weights for each parallel branch using global information, EMA further aggregates the output features of the two parallel branches through cross-dimensional interaction to capture pixel-wise correspondences.

3.2.3. iEMA module

iRMB is a hybrid architecture designed for lightweight models. At its core, iRMB decouples and reorganizes the structural and functional components of the traditional CNN-based Inverted Residual Block (IRB) with attention mechanisms. A comparison between iRB and iRMB is illustrated in the Figure 2.

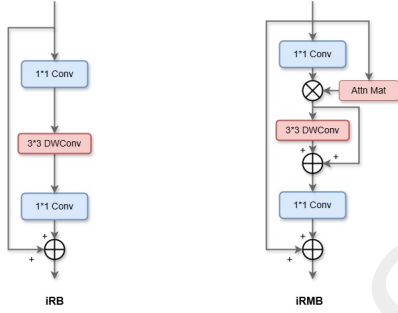


Figure 2: Comparison of iRB and iRMB.

iRMB employs a window-based spatial self-attention mechanism (Window-MHSA), which focuses on limited regions only. Although this design is efficient, it results in suboptimal performance for large or continuous objects and lacks cross-scale interaction and lead to diminished accuracy in small object detection. To address these limitations, iEMA builds upon the iRMB structure by replacing the window-based attention to the EMA mechanism. iEMA captures both fine-grained local details and global semantic context by incorporating multi-scale parallel branches (1×1 , 3×3 , and 5×5 convolutions). Additionally, it introduces a cross-dimensional interaction module to dynamically fuse features across different resolutions. This design can significantly reduce the miss rate for small objects and enhances feature extraction for large, cross-window targets.

3.2.4. EBP structure

This paper proposes a cross-layer feature extraction network structure and named EBP. This structure is inspired by the BiFPN architecture to extract features of small objects better.

For a further enhancement on the effectiveness for small object detection, the proposed EBP structure introduces an additional feature layer and integrates the iEMA module, it enables deeper and more efficient extraction of meaningful features from the image. The detailed by incorporating the EMA architecture is illustrated in the Figure 3.

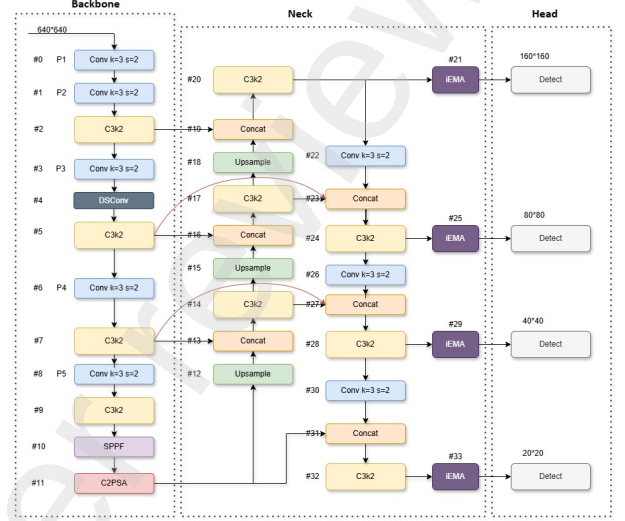


Figure 3: Architecture Diagram of ESW-YOLO.

Based on YOLO11, the EBP architecture introduces a P2-level detection head and enhances the richness of fine-grained features by integrating cross-layer feature fusion at the P3 and P4 levels. This design facilitates deeper feature extraction while preserving higher-resolution feature maps crucial for small object detection. The iEMA module further refines the incoming multi-channel feature maps by extracting the most informative elements and assigning appropriate weights to each channel. This structure not only enriches the feature representation for small object detection but also ensures efficient and effective use of these features, capturing more essential target characteristics while maintaining a lightweight footprint.

3.3. Wies-IoU

WIoU is an IoU-based loss function, quipped with a dynamic non-monotonic focusing mechanism (FM) that evaluates anchor box quality based on outlier degree rather than IoU. The calculation formula of its v3 version is as follows:

$$L_{WIoU} = rR_{WIoU}L_{IoU} \quad (9)$$

$$r = \frac{\beta}{\delta\alpha^{\beta} - \delta} \quad (10)$$

$$\beta = \frac{L_{IoU}^*}{L_{IoU}} \in [0, +\infty) \quad (11)$$

$$R_{WIoU} = \exp\left(\frac{(x - x_{gt}^2) + (y - y_{gt}^2)}{(W_g^2 + H_g^2)}\right) \quad (12)$$

Here, L_{WIoU} denotes the WIoU loss function, and r represents the gradient gain. R_{WIoU} is a penalty function. $\bar{L}_{IoU}$ indicate the mean value of loss function. β denotes the outlier degree, while δ and α are tunable hyperparameters used to modulate the value of r . L_{IoU}^* represents the current loss. The coordinates of the anchor box center are denoted as (x, y) , and (x_{gt}, y_{gt}) indicate the center of the ground truth box. Finally, W_g and H_g represent the width and height of the ground truth and anchor boxes, respectively. In this study, we adopt $r = 1$, which corresponds to WIoU v1.

4. experimental results and analysis

4.1. Dataset

4.1.1. PKU-Market-PCB Defect Dataset

The Printed Circuit Board (PCB) Defect Dataset is a publicly available synthetic dataset. It is designed for tasks such as image detection, classification, and registration. Notably, the defective regions are typically confined to areas of 12×12 pixels or smaller, making this dataset an excellent benchmark for evaluating the performance of algorithms in small object detection. The details of dataset are shown in the first line of Figure 4.

4.1.2. RSOD Remote sensing image dataset

RSOD is an open remote sensing object detection dataset comprising four categories, each stored in a separate folder. As shown in the second line of Figure 4, the targets in the dataset — airplanes, storage tanks, playgrounds, and overpasses — feature a large number of instances smaller than 1,024 pixels, making it a valuable benchmark for evaluating the generalization ability of detection models in this study.

4.2. Experimental Setup

The experiments were conducted using the PyTorch 2.6.0 deep learning framework with CUDA 12.4, and the programming language is Python 3.12.9. The system run on Windows 10, equipped with an i7-8750H processor and 16 GB of RAM, and using NVIDIA GeForce GTX 1060 with 6 GB of VRAM as GPU. The dataset was divided into training, testing, and validation sets in a ratio of 7:1:2.

4.3. Evaluation Metrics

TP refers to positive samples correctly predicted by the model, TN to negative samples correctly predicted, FP to negative samples incorrectly predicted as positive, and FN to positive samples incorrectly predicted as negative. Their relationships are summarized in Table 1.

Precision (P) quantifies the proportion of true positive samples among all samples predicted as positive. The calculation formula is as follows:

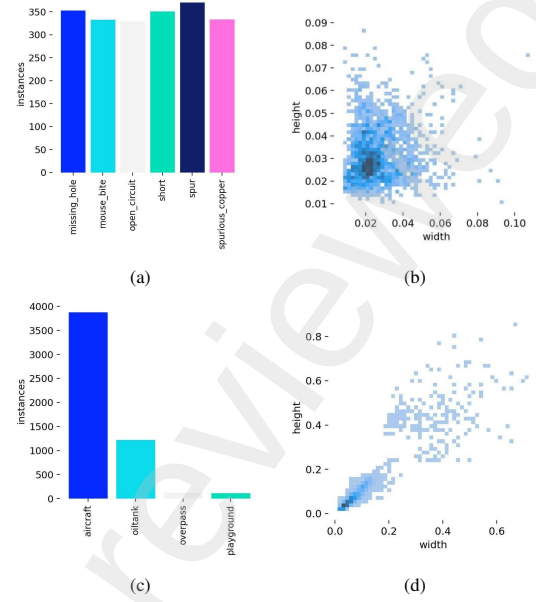


Figure 4: The quantity and size distribution of targets within the datasets. (a) illustrates the category distribution of targets in the PKU-Market-PCB dataset; (b) depicts the size distribution of targets in the PKU-Market-PCBdataset; (c) shows the category distribution of targets in the RSOD remote sensing image dataset; (d) presents the size distribution of targets in the RSOD remote sensing image dataset.

Ground Truth	Predicted Results	
	positive	negative
positive	TP(true positive)	FN(false negative)
negative	FP(false positive)	TN(true negative)

Table 1: The relationship of TP,FP,FN,TN

$$P = \frac{TP}{TP + FP} \quad (13)$$

Recall (R) measures the proportion of actual positive samples that are correctly predicted by the model. The calculation formula is as follows:

$$R = \frac{TP}{T + FN} \quad (14)$$

Mean Average Precision (mAP) is a widely used evaluation metric in object detection, measuring model accuracy across different thresholds. This study employs two mAP metrics: mAP50 and mAP50-95; mAP50: The mean average precision at an Intersection over Union (IoU) threshold of 0.50; mAP50-95: The mean average precision calculated across IoU thresholds ranging from 0.50 to 0.95, with an increment of 0.05; GFLOP (Giga Floating Point Operations per Second) represents the number of billion floating-point operations executed per second. It serves as a standard metric for assessing a model's computational complexity. Lower GFLOP values typically indicate higher computational efficiency on resource-constrained devices, whereas higher GFLOP values often re-

quire more advanced hardware to achieve real-time performance.

4.4. Comparison with Previous YOLO Versions

The first set of experiments primarily aims to validate the proposed model's effectiveness in small object detection. It conducted by 200 epochs using the PKU-Market-PCB dataset, where all samples are smaller than 12×12 pixels, making it an ideal benchmark to check its performance in SOD.

The experiment compared ESW-YOLO with YOLOv5s, YOLOv8n, YOLOv8s, YOLO11n, and YOLO11s. To clearly present the experimental results, the obtained metrics are summarized in Table 2. Figure 5 illustrates these models' mAP across 200 training epochs.

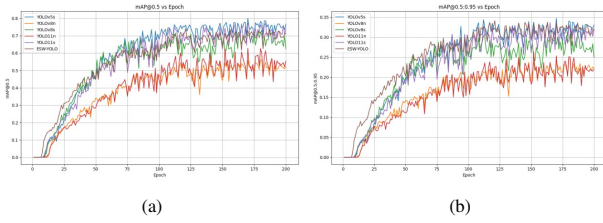


Figure 5: Performance comparison curves for the ESW-YOLO and YOLO series. (a) mAP0.5; (b) mAP0.5:0.95 value.

As the experimental data in Table 2, ESW-YOLO achieves a remarkable leap in detection accuracy, only sacrifices slightly in computational complexity. Specifically, on small objects, it boosts mAP50 by 12.7%, representing a 20.5% improvement over YOLO11n. Under the mAP50-95 evaluation, ESW-YOLO surpasses models like YOLOv8s and YOLOv5s, whose parameter counts are three to four times larger. ESW-YOLO achieving high-precision small-object detection while remaining lightweight.

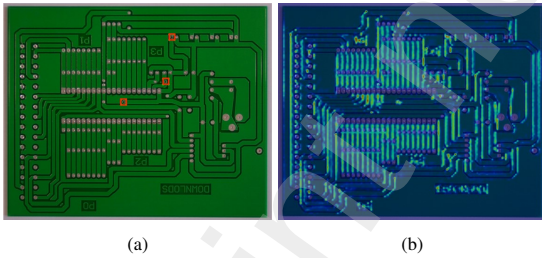


Figure 6: Heatmap of ESW-YOLO for "missing hole", targets are highlighted in the red box.

The targets in Figure 6 are "missing hole." Figure 6b shows the corresponding heatmap results, while Figure 6a presents the feature map from the P2 detection layer after processing with the EMA attention mechanism. It is evident that the target is prominently highlighted, further demonstrating the strong feature extraction capability of our model.

4.5. Ablation Experiment

To demonstrate the impact of each improvement and module on the model's performance, we conducted an ablation study on

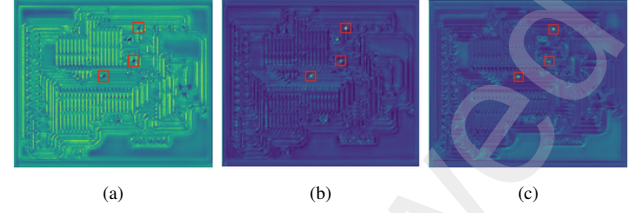


Figure 7: Feature maps extracted at the P2 level in ESW-YOLO. Targets in the input image are highlighted in red boxes.

ESW-YOLO. The experiment involved removing and reassembling individual components to analyze the contribution of each module. The results are shown in the Table 3.

It is evident that when targeting detection tasks for objects with pixel areas below 144, each improvement has contributed to better detection performance.

From the experimental results, it can be observed that when these modules is added to the base YOLO11n model individually, the impact of WIoU on mAP is only a 0.2% increase. By using EBP structure, module has a 6.5% increase in mAP, though the GLFOPs increased by 4.4. Conversely, DSConv, when applied independently, actually degraded the system's performance. This is because, under the base structure, the system struggled to effectively utilize the deep features extracted by DSConv, leading to the additional computational cost adversely affecting model accuracy.

When DSConv is combined with WIoU, the model's mAP shows a slight improvement compared to using DSConv alone. This indicates that the features extracted by DSConv are more effectively utilized when WIoU is used as the loss function, compared to using YOLO11's original CIoU loss function. The result of using both DSConv and EBP together is similar to using EBP alone, which suggests that the features extracted by DSConv have not been fully exploited. However, when all three modules are used together, the experimental results show a significant improvement in both recall and mAP, outperforming all other structures. This is because the combination of EBP and WIoU facilitates better utilization of the feature information extracted by DSConv. The EBP structure has a stronger capability in feature extraction and weight allocation for feature maps.

4.6. Application on Other Datasets

To verify the generalization ability of our proposed model in small object detection, we selected the RSOD remote sensing image dataset for the generalization experiment. The epoch was set to 200, and the accuracy of models on the same performance level was observed. Figure 9 presents a heatmap feature analysis comparison of the four models.

Table 4 presents the experimental results of YOLOv5s, YOLOv8s, YOLO11s, and ESW-YOLO on the RSOD remote sensing image dataset. The data clearly demonstrate that our model achieves detection accuracy on par with, or even exceeding, models that are 3-4 times larger in parameter size. ESW-YOLO delivers the best performance over other module under the mAP0.5 metric, and can fully meets the requirements for

Models	Precision/%	Recall/%	mAP0.5/%	mAP0.5:0.95/%	Parameter/ 10^6	GFLOP
YOLOv5s	0.919	0.703	0.801	0.348	9.113858	23.8
YOLOv8n	0.679	0.533	0.559	0.246	3.006818	8.1
YOLOv8s	0.737	0.703	0.716	0.309	11.127906	28.4
YOLOv11n	0.772	0.568	0.619	0.264	2.583322	6.3
YOLO11s	0.896	0.691	0.766	0.336	9.415122	21.3
ESW-YOLO	0.812	0.647	0.746	0.352	2.816122	11.1

Table 2: Comparison Between YOLO Family Representatives and ESW-YOLO

Baseline	WIoUv1	EBP	DSConv	Precision/%	Recall/%	mAP0.5/%	mAP0.5:0.95/%	Parameter/ 10^6	GFLOP
YOLO11n				0.772	0.568	0.619	0.264	2.583322	6.3
	✓			0.645	0.607	0.621	0.263	2.583322	6.3
		✓		0.806	0.614	0.684	0.326	2.778294	10.7
			✓	0.624	0.562	0.574	0.238	2.621150	6.7
		✓	✓	0.852	0.558	0.685	0.322	2.816122	11.1
	✓		✓	0.776	0.544	0.59	0.254	2.621150	6.7
	✓	✓		0.803	0.641	0.683	0.309	2.778294	10.7
	✓	✓	✓	0.812	0.647	0.746	0.352	2.816122	11.1

Table 3: Ablation Experiment of ESW-YOLO.

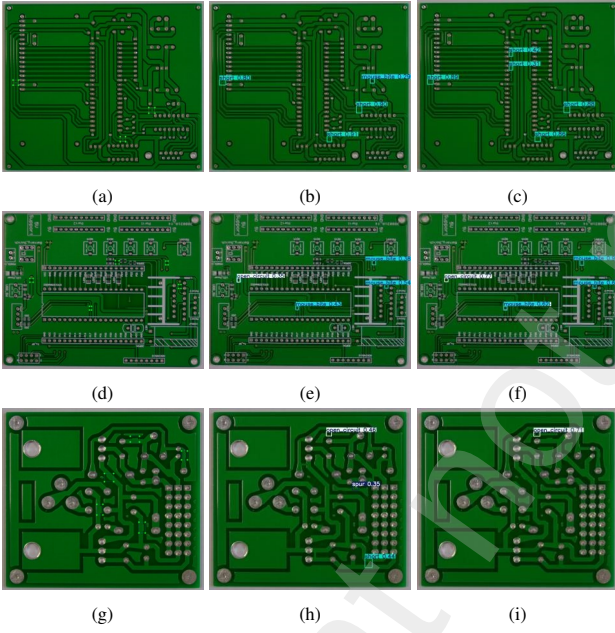


Figure 8: The first column displays the input images, the second column presents the detection results of YOLO11n, the third column shows the results of ESW-YOLO without DSConv, and the fourth column illustrates the detection outcomes of ESW-YOLO. The performance and comparisons are shown for the mouse bite, open circuit, and short defects.

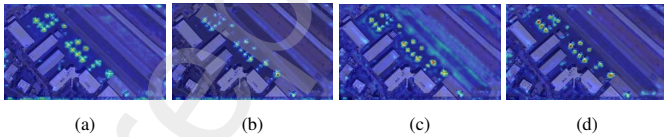


Figure 9: The first column displays the input images, the second column presents the detection results of YOLO11n, the third column shows the results of ESW-YOLO without DSConv, and the fourth column illustrates the detection outcomes of ESW-YOLO. The performance and comparisons are shown for the mouse bite, open circuit, and short defects.

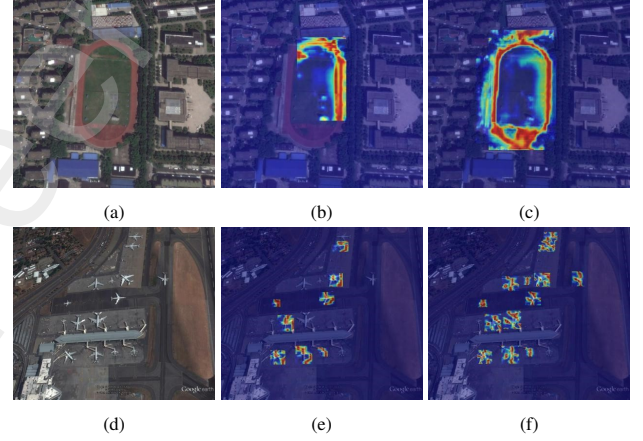


Figure 10: The first column shows the input, the second column displays the heatmap features obtained from the baseline, and the third column shows the heatmap features of ESW-YOLO in remote sensing small target recognition.

real-time detection. Additionally, its lower GFLOPs make it more adaptable to a wider range of hardware platforms.

5. Conclusion

The proposed ESW-YOLO achieves a remarkable 20.5% improvement in mAP compared to the original YOLO11n under the same parameter scale, while reducing the model size to less than one-third of other YOLO variants with comparable accuracy. Through the integration of the well-designed EBP structure, the advanced EMA attention mechanism, the efficient WIoU loss function, and the small-object-enhanced DSConv module, ESW-YOLO effectively elevates the performance of lightweight models. Moreover, the incorporation of the DWConv3k2 module further lightens the model, offering a flexible trade-off between model compactness and detection performance.

Models	Precision/%	Recall/%	mAP0.5/%	mAP0.5:0.95/%	Parameter/10 ⁶	GFLOP
YOLOv5s	0.514	0.54	0.508	0.342	9,113,084	23.8
YOLOv8s	0.415	0.657	0.504	0.343	11,127,132	28.4
YOLO11s	0.524	0.514	0.509	0.334	9,414,348	21.3
ESW-YOLO	0.52	0.529	0.525	0.337	2,815,858	11.1

Table 4: Comparison of the performance of ESW-YOLO, YOLOv5s, YOLOv8s, and YOLO11s on the RSOD Remote Sensing Image Dataset.

However, despite these achievements, there is still room for improvement in the model's detection speed. Although ESW-YOLO meets the requirements for real-time detection and maintains relatively low computational overhead, it falls short in delivering higher inference speed.

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